

MATH 2211, SPRING 2026: HOMEWORK 8

- (1) Show that the following are equivalent for a matrix $A \in M_{n \times n}(F)$ and a scalar $\lambda \in F$:
- (a) There exists a non-zero vector $v \in F^n$ such that $Av = \lambda v$.
 - (b) The RREF of $A - \lambda I_n$ has a free variable.
 - (c) If $T : F^n \xrightarrow{v \mapsto Av} F^n$ is the linear transformation associated with A , then $T - \lambda \cdot \text{id}_{F^n}$ is not injective.
- (2) Prove that the determinant of 2×2 -matrices has the following property:

$$\det(AB) = \det(A) \det(B)$$

for all $A, B \in M_2(F)$.

We will see in class that there is a *unique* function $\det : M_n(F) \rightarrow F$ with the following properties:

(a) (Identity) $\det(I_n) = 1$.

(b) (Alternating property) $\det \begin{pmatrix} v_1^T \\ v_2^T \\ \vdots \\ v_n^T \end{pmatrix} = 0$;

(c) (Scaling) $\det \begin{pmatrix} c \cdot v_1^T \\ v_2^T \\ \vdots \\ v_n^T \end{pmatrix} = c \cdot \det \begin{pmatrix} v_1^T \\ v_2^T \\ \vdots \\ v_n^T \end{pmatrix}$

(d) (Addition) $\det \begin{pmatrix} v_1^T + w_1^T \\ v_2^T \\ \vdots \\ v_n^T \end{pmatrix} = \det \begin{pmatrix} v_1^T \\ v_2^T \\ \vdots \\ v_n^T \end{pmatrix} + \det \begin{pmatrix} w_1^T \\ v_2^T \\ \vdots \\ v_n^T \end{pmatrix}$

- (3) Show that $\det$ also has the following properties:

(a) (Swapping property) $\det \begin{pmatrix} v_1^T \\ v_2^T \\ \vdots \\ v_i^T \\ v_{i+1}^T \\ \vdots \\ v_n^T \end{pmatrix} = -\det \begin{pmatrix} v_1^T \\ v_2^T \\ \vdots \\ v_{i+1}^T \\ v_i^T \\ \vdots \\ v_n^T \end{pmatrix}$

(b) (Adding rows property) $\det \begin{pmatrix} \begin{bmatrix} v_1^T \\ v_2^T \\ \vdots \\ v_i^T \\ v_{i+1}^T \\ \vdots \\ v_n^T \end{bmatrix} \end{pmatrix} = \det \begin{pmatrix} \begin{bmatrix} v_1^T \\ v_2^T \\ \vdots \\ v_i^T + v_{i+1}^T \\ v_{i+1}^T \\ \vdots \\ v_n^T \end{bmatrix} \end{pmatrix}$

- (4) For any matrix $B \in M_n(F)$, show that the function $A \mapsto \det(AB)$ also has the defining properties (b), (c), (d) of $\det$, except that it takes I_n to $\det(B)$. Use this to show that we have $\det(AB) = \det(A) \det(B)$ even in the $n \times n$ -case.

- (5) Use row operations to compute $\det(A)$ where

$$A = \begin{bmatrix} 1 & 2 & -1 & 0 & 1 \\ 2 & 5 & 0 & 2 & 3 \\ -1 & -2 & 4 & 1 & 0 \\ 0 & 3 & 6 & 11 & 5 \\ 1 & 4 & 1 & 5 & 7 \end{bmatrix}$$